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EX:2 write a python program to solve 8-Queen problem

AIM

To write a Python program to solve the 8-Queen Problem using the Backtracking Algorithm, placing 8 queens on a chessboard so that no two queens attack each other.

ALGORITHM

1. Create an empty 8×8 chessboard.
2. Start placing a queen from the first row.
3. Check whether the selected position is safe.
4. Check the same column and both upper diagonals.
5. If safe, place the queen and move to the next row.
6. If no safe position is available, use backtracking and try another position.
7. Repeat until all 8 queens are placed.
8. Display the final solution.

PROGRAM CODE:

```
N = 8
```

```
board = [[0 for _ in range(N)] for _ in range(N)]
```

```
def is_safe(row, col):
```

```
    # Check the same column
```

```
    for i in range(row):
```

```
        if board[i][col] == 1:
```

```
            return False
```

```
    # Check upper-left diagonal
```

```
i, j = row - 1, col - 1
```

```
while i >= 0 and j >= 0:
```

```
    if board[i][j] == 1:
```

```
        return False
```

```
    i -= 1
```

```
    j -= 1
```

```
# Check upper-right diagonal
```

```
i, j = row - 1, col + 1
```

```
while i >= 0 and j < N:
```

```
    if board[i][j] == 1:
```

```
        return False
```

```
    i -= 1
```

```
    j += 1
```

```
return True
```

```
def solve_queen(row):
```

```
    if row == N:
```

```
        return True
```

```
    for col in range(N):
```

```
        if is_safe(row, col):
```

```
            board[row][col] = 1
```

```

        if solve_queen(row + 1):

            return True

        board[row][col] = 0

    return False

if solve_queen(0):

    print("8-Queen Solution:")

    for row in board:

        print(" ".join("Q" if x == 1 else "." for x in row))

else:

    print("No solution found.")

```

OUTPUT

```

=====
8-Queen Solution:
Q . . . . . . .
. . . . Q . . .
. . . . . . Q
. . . . Q . . .
. . Q . . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . . .
>>> |

```

RESULT

Thus, the 8-Queen Problem was successfully solved using the Backtracking Algorithm in Python, with all 8 queens placed so that no two queens attack each other.